Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (1 pt)**.** Show that if G is a connected graph that is not regular, then G contains **adjacent** vertices u and v such that $\deg u \neq \deg v$.

Hint: there is a reason why this problem is worth 1 pt.

Solution. Consider the function $\deg\colon V(G) \to \mathbb{Z}$ sending a vertex v in G to $\deg v$. The fact that G is not regular just means that this function is not constant. Thus, since G is connected, by HW2.2, we know that there there is an edge uv in G such that $\deg u \neq \deg v$. □

Problem 2 (2 pts)**.** Let G be a graph on n vertices. Prove that if $\Delta(G) + \delta(G) \geqslant n - 1$ then G is connected.

Solution. Fix a vertex v^* in G such that $\deg v^* = \Delta(G)$. To show that G is connected, it suffices to show that there is a v^*–v walk for any other vertex v in G. Notice that

$$\deg v^* + \deg v = \Delta(G) + \deg v \geqslant \Delta(G) + \delta(G) \geqslant n - 1.$$

Now we consider two cases. If $v^*v \in E(G)$, we clearly have a walk. So we assume that v^* and v are not adjacent. Since $\deg v^* + \deg v \geqslant n - 1$, we saw in class (see the proof of Theorem 2.4) that there is some $w \in N(v^*) \cap N(v)$. Thus (v^*, w, v) is a v^*–v walk. □

Solution. (Alternative) Let us show the contrapositive. Assume G is not connected, thus it has $k \geqslant 2$ components $G_1, \ldots, G_k$ such that $G = G_1 \sqcup \cdots \sqcup G_k$. Let $n_i = |V(G_i)$ thus $n = n_1 + \cdots + n_k$. Without loss of generality we can assume $n_1 \geqslant \cdots \geqslant n_k$. Observe that

$$\Delta(G) = \max\{\Delta(G_i)\colon i \in \{1, \ldots, k\}\} \leqslant \max\{n_i - 1\colon i \in \{1, \ldots, k\}\} = n_1 - 1,$$
$$\delta(G) = \min\{\delta(G_i)\colon i \in \{1, \ldots, k\}\} \leqslant \min\{n_i - 1\colon i \in \{1, \ldots, k\}\} = n_k - 1.$$

Thus $\Delta(G) + \delta(G) \leqslant n_1 - 1 + n_k - 1 \leqslant n - 2$ □

Problem 3 (2 pts)**.** Let G be a graph with the property that $\deg u + \deg v \equiv 1 \pmod 2$ for every $uv \in E(G)$. Prove that $|E(G)|$ is even.

Solution. For $i = 0, 1$, consider the set $A_i = \{v \in V(G)\colon \deg v \equiv i \pmod 2\}$, thus $V(G) = A_0 \sqcup A_1$. By hypothesis any edge in G connects two vertices with different parities, thus A_0 and A_1 and independent sets and the partition $V(G) = A_0 \sqcup A_1$ is actually a bipartition. Now we can apply HW 2.5 to compute

$$|E(G)| = \sum_{v \in A_0} \deg v,$$

which must be even because its a sum of even numbers. □

Problem 4 (2 pts)**.** Let G be a connected graph and suppose that $f\colon V(G) \to \mathbb{Z}$ is a function with the property that $f(u) + f(v) \equiv 0 \pmod 3$ for every $uv \in E(G)$. Prove that if G is *not* bipartite, then $f(v) \equiv 0 \pmod 3$ for every $v \in V(G)$.

Solution. Let us show the contrapositive. So we assume there is some $v^* \in V(G)$ such that $f(v^*) \not\equiv 0 \pmod 3$, and want to show that G is bipartite.

For $i = 0, 1, 2$, consider the set $A_i = \{v \in V(G) \colon \deg v \equiv i \pmod 3\}$, thus $V(G) = A_0 \sqcup A_1 \sqcup A_2$. Observe that there are no edges between vertices in A_1, because $1 + 1 \equiv 1 \pmod 3$, thus A_1 is independent. Similarly, A_2 is independent because $2 + 2 \equiv 1 \pmod 3$. Thus, if we manage to show that A_0 is empty, then we obtain a bipartition $V(G) = A_1 \sqcup A_2$.

Suppose for a contradiction that $A_0 \neq \varnothing$ and consider $B = A_1 \cup A_2$ which contains v^*. Since G is connected and $V(G) = A_0 \sqcup B$ with A and B nonempty, there must be an edge ab with $a \in A_0$ and $b \in B$. But this edge satisfies $f(a) + f(b) \equiv f(b) \pmod 3 \not\equiv 0 \pmod 3$, a contradiction. □

Problem 5 (2 pts)**.** Let G be a graph on n vertices. Prove the following:

1. $\delta(G) + \delta(\overline{G}) \leqslant n - 1$,
2. $\delta(G) + \delta(\overline{G}) = n - 1$ if and only if G is regular.

Solution. 1. Observe that for any vertex we have $\deg_{\overline{G}} v = n - 1 - \deg_G v$, thus

$$\delta(\overline{G}) = \min\{\deg_{\overline{G}} v \colon v \in V(\overline{G})\} = n - 1 - \max\{\deg_G v \colon v \in V(G)\} = n - 1 - \Delta(G)$$

Since $\delta(G) - \Delta(G) \leqslant 0$ we have

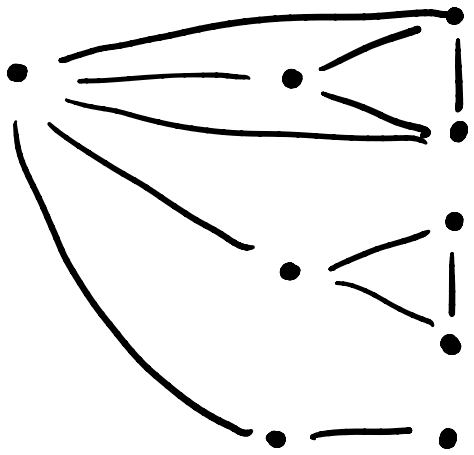
$$\delta(G) + \delta(\overline{G}) = \delta(G) + n - 1 - \Delta(G) = n - 1 + (\delta(G) - \Delta(G)) \leqslant n - 1.$$

2. Using the argument above, $\delta(G) + \delta(\overline{G}) = n - 1$ if and only if $\delta(G) - \Delta(G) = 0$, which happens if and only if G is regular. □

Problem 6 (0.5+0.5 pts)**.** Determine whether or not the following sequences are graphical. If the sequence is graphical, draw a picture of a graph with that degree sequences. If the sequence is not graphical, prove this.

1. $5, 3, 3, 3, 3, 2, 2, 2, 1$
2. $6, 5, 5, 4, 3, 2, 1$

Solution. 1.This sequence is graphical, below we have a realization



2. We repeatedly apply Havel-Hakimi's theorem to produce

$$(6, 5, 5, 4, 3, 2, 1) \mapsto (4, 4, 3, 2, 1, 0) \mapsto (3, 2, 1, 0, 0) \mapsto (1, 0, 0, -1).$$

This last sequence is not graphical, so the original sequence is not graphical either. □